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Studierichting: Informatica
Oefeningenreeks 1E nr 24

$z^3 + az^2 + bz - 12$ deelbaar door $z - 3$ en $z - 2i$

$z(3) = 0 \Rightarrow 27 + 3a + 3b - 12 = 0$

$z(2i) = 0 \Rightarrow 8i^3 + 4i^2 + 2ib - 12 = -8i - 4a + 2ib - 12 = 0$

$$\left\{ \begin{array}{lcl} 3b & = & -9a - 15 \\ -8i - 4a + 2ib - 12 + 0 & = & 0 \end{array} \right.$$

$$\Leftrightarrow \left\{ \begin{array}{lcl} b & = & -5 - 3a \\ -8i - 4a - 10i - 6ai - 12 & = & 0 \end{array} \right.$$

$$\Leftrightarrow \left\{ \begin{array}{lcl} b & = & -5 - 3a \\ 18i - 12 & = & 4a + 6ai \end{array} \right.$$

$$\Leftrightarrow \left\{ \begin{array}{lcl} b & = & -5 - 3a \\ a & = & -3 \end{array} \right.$$

$$\Leftrightarrow \left\{ \begin{array}{lcl} b & = & 4 \\ a & = & -3 \end{array} \right.$$

References